

Supplementary Material

This supplementary file provides the full training configuration and statistical significance analysis for action classification referenced in the main manuscript.

I. FULL TRAINING DETAILS

During training, ATP-CLIP uses the joint objective defined in Section V-C. For base training, a physical batch size of 8 is adopted, with gradient accumulation to achieve an effective batch size of 32. For few-shot classification and cross-modal retrieval, the results are averaged in 5 independent trials with different random seeds and reported as *Mean $\pm$ Standard Deviation*. For clarity and auditability, we provide a comprehensive technical summary of our architectural configurations, optimization hyperparameters, and experimental protocols in Table S1.

II. STATISTICAL SIGNIFICANCE ANALYSIS FOR ACTION CLASSIFICATION

We conducted a statistical significance analysis to determine whether the classification gains of ATP-CLIP(L) were greater than the variation produced by repeated model training and data splitting. Reporting the mean and standard deviation over several runs was useful because it described the center and spread of the observed scores, but it did not directly test the difference between two methods. Therefore, we adopted the five seed-matched runs as the experimental units. For each dataset, ATP-CLIP(L) and the baseline were trained and evaluated with the same split and seed setting. This design produced five pairs of observations for each available comparison. Pairing was important in this setting because a given split could raise the scores of both methods. Comparing matched runs removed much of this shared split effect from the error term and focused the analysis on the within-split performance difference. For split i , the paired difference was defined as ATP-CLIP(L) minus the corresponding baseline score, defined by:

$$d_i = y_{\text{ATP},i} - y_{\text{B},i}, \quad i = 1, \dots, n, \quad n = 5. \quad (1)$$

A positive value of d_i in Equation (1) indicated that ATP-CLIP(L) achieved a higher score on that split, while a negative value indicated that the baseline achieved a higher score. The parameter tested by the paired procedure was the paired difference of a population mean, denoted by μ_d . The null hypothesis was $H_0: \mu_d = 0$, and the two-sided alternative was $H_1: \mu_d \neq 0$. A two-sided test was selected because it allowed the data to support either direction and did not assume in advance that ATP-CLIP(L) must be better. The sample mean difference and sample standard deviation of the five paired

differences summarized the estimated gain and its run-to-run variation. They were computed by:

$$\bar{d} = \frac{1}{n} \sum_{i=1}^n d_i, \quad (2a)$$

$$s_d = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (d_i - \bar{d})^2}. \quad (2b)$$

The paired t statistic divided the mean paired difference by its estimated standard error. With $n = 5$ paired runs, every comparison had four degrees of freedom. The raw p -value, denoted by p_{raw} , was obtained from the two-sided t distribution. A small p_{raw} meant that a mean difference at least as far from zero as the observed one would be unusual under the null model. It should not be interpreted as the probability that the null hypothesis was true. The paired t -test assumed that the paired differences were independent across runs and approximately normally distributed. Matching seeds and splits supported the paired construction, although the small number of runs limited the extent to which the distributional assumption could be tested. The test statistic, degrees of freedom, and raw p -value were computed by:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}, \quad (3a)$$

$$df = n - 1 = 4, \quad (3b)$$

$$p_{\text{raw}} = 2[1 - F_{t_4}(|t|)], \quad (3c)$$

To complement the p -values, we reported the unadjusted 95% confidence interval for the mean paired difference Δ . The interval used the same paired standard error and the critical value of the t distribution with four degrees of freedom. Because $t_{0.975,4} = 2.776$, the interval was wider than a large-sample normal interval. The interval gave the range of mean performance differences compatible with the five paired runs under the model. An interval entirely above zero supported a positive mean difference before multiple-comparison correction. An interval containing zero indicated that the available runs were also compatible with no mean difference. The confidence limits were expressed in percentage points, so their magnitude could be interpreted directly as a performance margin. The interval was computed by:

$$\text{CI}_{95\%}(\Delta) = \bar{d} \pm t_{0.975,4} \frac{s_d}{\sqrt{n}}, \quad t_{0.975,4} = 2.776. \quad (4)$$

Multiple baselines were evaluated within each dataset and metric, creating a multiple-testing problem. If every raw p -value were judged independently at 0.05, the probability of obtaining at least one false-positive result would increase

TABLE S1

COMPREHENSIVE TRAINING SUMMARY OF ATP-CLIP. WE DETAIL THE ARCHITECTURE DIMENSIONS, OPTIMIZATION STATUS, AND CORE TRAINING HYPERPARAMETERS. $D \in \{512, 768, 1024\}$ DENOTES THE FEATURE DIMENSIONS OF SMALL, MEDIUM, AND LARGE VARIANTS.

Category	Component/Parameter	Value/Detail	Status
<i>Backbone</i>	Visual Encoder (CLIP ViT)	$T \times D$	Frozen
	Text Encoder (CLIP Trans.)	$1 \times D$	Frozen
	Checkpoint Provenance	<i>OpenAI (WIT)</i> $\rightarrow$ <i>K400</i>	–
<i>Adapters</i>	TAP Module (Conv1D + Attn)	$1 \times D$	<i>Trainable</i>
	W-Adapter (MLP, r=4)	$D \rightarrow D/4 \rightarrow D$	<i>Trainable</i>
	Classification Head	$1 \times C$	<i>Trainable</i>
<i>Training</i>	Optimizer	AdamW	–
	Learning Rate / Schedule	$1e^{-4}$ / Cosine Anneal	–
	Batch Size (Physical / Eff.)	8 / 32 (Grad Accum)	–
	Regularization	Dropout (0.2), Label Smooth (0.1)	–
	Boss Balancing (λ)	0.5	–
<i>Protocol</i>	Few-shot Evaluation	Base-to-Novel Split	–
	Random Seed Handling	5 Independent Trials	–
	Subject Constraints	Strict Subject-Disjoint	–

with the number of comparisons. We applied the Holm step-down procedure separately within each dataset–metric family. Unavailable comparisons were not included. The family sizes were eight for SSBD Top-1, five for SSBD Macro-F1, six for ESBD Top-1, and four for ESBD Macro-F1. Within each family, the raw p -values were ordered from smallest to largest. The Holm-adjusted value for the k th ordered test was computed using the following monotone step-down rule by:

$$p_{H,(k)} = \max_{1 \leq j \leq k} \min \{1, (m - j + 1)p_{(j)}\}, \quad (5)$$

$$k = 1, \dots, m.$$

The Holm procedure in Equation (5) controlled the family-wise error rate while retaining more power than applying the same Bonferroni threshold to every test. Our final significance decision was based on $p_H < 0.05$, rather than on p_{raw} alone. The star in Table S2 was therefore attached only to the Holm-adjusted p -value. The reported confidence intervals (CIs) were unadjusted, so a 95% interval could exclude zero even when the corresponding comparison did not remain significant after Holm correction. This occurred for the SSBD Top-1 comparison with VideoMAE. There was no conflict between the interval and the adjusted test. The two quantities answered questions at different correction levels. Across all four families, 15 of the 23 available comparisons remained significant after correction.

On SSBD Top-1 accuracy, ATP-CLIP(L) showed clear corrected improvements over the two lower-scoring visual baselines and all three CLIP-based baselines. The comparison with 3D-CNN+TCN had $p_H = 0.0184$ and a 95% confidence interval of [2.99, 8.61]. The comparison with LSTM+MobNet had $p_H = 0.0011$ and a much larger interval of [14.95, 21.45]. ATP-CLIP(L) also significantly outperformed CLIP, VideoCLIP, and ActionCLIP, with adjusted p -values of 0.0022, 0.0040, and 0.0062, respectively. Their CIs were [7.56, 12.24], [5.65, 10.35], and [4.37, 8.83], and every lower limit was above

zero. These results showed that the improvement over the conventional CLIP-based methods was not driven by one favorable split. By contrast, the comparisons with Two-stage TAL and Swin-T+Lang were not significant. Their estimated margins were smaller, and their intervals, $[-0.60, 4.40]$ and $[-1.08, 2.68]$, included zero. The raw SSBD Top-1 comparison with VideoMAE was significant at $p_p = 0.0245$ and had an unadjusted interval of [0.76, 6.44]. After accounting for the eight comparisons in this family, however, p_H increased to 0.0734. We described the observed VideoMAE gain as a positive numerical result that did not meet the corrected significance criterion.

The SSBD Macro-F1 results followed a similar pattern, although fewer visual baselines reported this metric. ATP-CLIP(L) significantly exceeded LSTM+MobNet with $p_H = 0.0005$ and a 95% interval of [16.66, 22.54]. It also significantly exceeded CLIP, VideoCLIP, and ActionCLIP. Their adjusted p -values were 0.0164, 0.0276, and 0.0276, while the corresponding CIs were [3.08, 8.52], [1.85, 7.15], and [1.32, 6.48]. The positive lower limits indicated that the average Macro-F1 gain was consistently above zero in these comparisons. The difference from VideoMAE was not significant after correction, with $p_H = 0.0706$ and an interval of $[-0.38, 5.98]$. This interval contained both a very small VideoMAE advantage and a meaningful ATP-CLIP(L) advantage. The five-run evidence therefore did not estimate this difference precisely enough to support a directional claim.

On ESBD Top-1 accuracy, the corrected evidence was again consistent for the three CLIP-based baselines. ATP-CLIP(L) significantly outperformed CLIP, VideoCLIP, and ActionCLIP with p_H values of 0.0035, 0.0075, and 0.0083. The respective 95% CIs were [6.03, 10.77], [4.23, 8.97], and [3.47, 7.94] percentage points. These intervals showed gains of similar scale to those observed on SSBD and supported the view that ATP-CLIP(L) improved the standard CLIP-based classification

TABLE S2
 PAIRED STATISTICAL COMPARISONS BETWEEN ATP-CLIP(L) AND EXTERNAL BASELINES FOR ACTION CLASSIFICATION ON SSBD AND ESB. D.

Method	SSBD						ESBD					
	Top-1			Macro-F1			Top-1			Macro-F1		
	p_{raw}	p_H	$CI_{95\%}(\Delta)$	p_{raw}	p_H	$CI_{95\%}(\Delta)$	p_{raw}	p_H	$CI_{95\%}(\Delta)$	p_{raw}	p_H	$CI_{95\%}(\Delta)$
<i>Visual baselines</i>												
3D-CNN+TCN	0.0046	0.0184*	[2.99, 8.61]	—	—	—	—	—	—	—	—	—
LSTM+MobNet	0.0001	0.0011*	[14.95, 21.45]	< 0.0001	0.0005*	[16.66, 22.54]	—	—	—	—	—	—
Two-stage TAL	0.1021	0.2041	[−0.60, 4.40]	—	—	—	0.3104	0.3104	[−1.39, 3.39]	—	—	—
Swin-T+Lang	0.3037	0.3037	[−1.08, 2.68]	—	—	—	0.0631	0.1894	[−3.55, 0.15]	—	—	—
VideoMAE	0.0245	0.0734	[0.76, 6.44]	0.0706	0.0706	[−0.38, 5.98]	0.0879	0.1894	[−0.57, 5.37]	0.1074	0.1074	[−0.82, 5.62]
<i>CLIP-based methods</i>												
CLIP	0.0003	0.0022*	[7.56, 12.24]	0.0041	0.0164*	[3.08, 8.52]	0.0006	0.0035*	[6.03, 10.77]	0.0036	0.0144*	[3.06, 8.14]
VideoCLIP	0.0007	0.0040*	[5.65, 10.35]	0.0092	0.0276*	[1.85, 7.15]	0.0015	0.0075*	[4.23, 8.97]	0.0090	0.0271*	[1.83, 6.97]
ActionCLIP	0.0012	0.0062*	[4.37, 8.83]	0.0137	0.0276*	[1.32, 6.48]	0.0021	0.0083*	[3.47, 7.94]	0.0172	0.0345*	[1.05, 6.15]

p_{raw} denotes the raw p -value from a two-sided paired t -test over five seed-matched runs, and p_H denotes the corresponding Holm-adjusted p -value, with correction performed separately within each dataset–metric family. $CI_{95\%}(\Delta)$ denotes the unadjusted 95% confidence interval for the mean paired performance difference, $\Delta = \text{ATP-CLIP(L)} - \text{baseline}$, expressed in percentage points. The symbol * indicates statistical significance after Holm correction ($p_H < 0.05$); no significance marker is attached to the raw p_{raw} values. The symbol “—” denotes an unavailable comparison.

pipeline across datasets. The comparisons with Two-stage TAL, Swin-T+Lang, and VideoMAE did not reach corrected significance. Two-stage TAL had $p_H = 0.3104$ and an interval of $[-1.39, 3.39]$. VideoMAE had $p_H = 0.1894$ and an interval of $[-0.57, 5.37]$. In both cases, ATP-CLIP(L) had a higher reported mean, but the paired intervals were wide enough to include zero. Swin-T+Lang had a numerically higher mean Top-1 score than ATP-CLIP(L). This produced a negative mean paired difference, but the interval $[-3.55, 0.15]$ still included zero and p_H was 0.1894. The appropriate conclusion was that these methods were not statistically separated by the present five-run experiment, rather than that one method had been shown to be equivalent to the other.

For ESB. D Macro-F1, ATP-CLIP(L) significantly improved on all three CLIP-based methods. The comparison with CLIP yielded $p_H = 0.0144$ and an interval of $[3.06, 8.14]$. The comparison with VideoCLIP yielded $p_H = 0.0271$ and an interval of $[1.83, 6.97]$, while the comparison with ActionCLIP yielded $p_H = 0.0345$ and an interval of $[1.05, 6.15]$. The lower confidence limits remained positive even for the closest of these comparisons. VideoMAE was the only available visual baseline for ESB. D Macro-F1. ATP-CLIP(L) achieved the higher mean score, but the comparison was not significant, with $p_H = 0.1074$ and an interval of $[-0.82, 5.62]$. This result again indicated a positive point estimate with uncertainty that was too large to exclude a small difference in the opposite direction.

Several factors could explain why a numerical gain did not become statistically significant, and the result should not automatically be viewed as a failure of the proposed model. First, the strongest visual baselines already achieved high classification accuracy, so the remaining performance margin was small. A small mean difference required low run-to-run variation or more paired runs to be estimated precisely. Second, the Holm correction deliberately raised the evidence

requirement when several baselines were tested in the same family. This protected the response from selective claims, but it could turn a raw significant result into a corrected non-significant result, as observed for VideoMAE on SSBD Top-1. Third, the effect of split composition could remain important even after pairing. Some splits might contain more difficult subjects, fewer examples of a behavior, or a different balance of behavior classes. Pairing reduced the variation of the shared split, but it could not remove a true interaction in which two models responded differently to the same split. Finally, a non-significant test meant that the data did not provide enough evidence to reject zero mean difference. It did not prove that the methods had identical performance. Demonstrating practical equivalence would require a predefined equivalence margin and a separate equivalence test.

The sizes of SSBD and ESB. D also affected the interpretation. Both datasets were small relative to general video-recognition benchmarks, and the number of subjects and behavior instances was limited. Consequently, changing the training and test split could produce noticeable changes in class balance, subject difficulty, and example diversity. These changes could increase the variance of Top-1 and Macro-F1 across runs. The paired design was useful because it compared the models under exactly the same split conditions, yet the statistical sample size was still five paired runs, not the number of videos or frames. With four degrees of freedom, CIs were necessarily wide unless the within-pair differences were highly stable. The small datasets therefore influenced the analysis indirectly by making run-level performance more sensitive to the selected subjects and examples. They also limited how broadly the estimated differences could be generalized beyond the current evaluation protocols.

Altogether, the analysis supported the conclusion that ATP-CLIP(L) reliably outperformed CLIP, VideoCLIP, and Action-CLIP, as all 12 comparisons across the two datasets and two

classification metrics remained significant after Holm correction, with every corresponding confidence interval entirely above zero. The evidence relative to the visual baselines was more selective. The large SSBD gains over 3D-CNN+TCN and LSTM+MobNet were statistically supported, whereas the differences from Two-stage TAL, Swin-T+Lang, and VideoMAE were generally smaller and were not resolved by five runs after correction. This distinction did not weaken the reported mean results. Instead, it separated stable improvements from comparisons that required more evidence. We therefore restricted claims of statistical superiority to the starred comparisons in Table S2 and described the remaining strong visual baselines as statistically indistinguishable under the present experimental setting.